



# Multiscale Feature Extraction and Fusion of Image and Text in VQA

Siyu Lu<sup>1</sup> · Yueming Ding<sup>1</sup> · Mingzhe Liu<sup>2</sup> · Zhengtong Yin<sup>3</sup> · Lirong Yin<sup>4</sup> · Wenfeng Zheng<sup>1</sup> 

Received: 1 November 2022 / Accepted: 31 March 2023  
© The Author(s) 2023

## Abstract

The Visual Question Answering (VQA) system is the process of finding useful information from images related to the question to answer the question correctly. It can be widely used in the fields of visual assistance, automated security surveillance, and intelligent interaction between robots and humans. However, the accuracy of VQA has not been ideal, and the main difficulty in its research is that the image features cannot well represent the scene and object information, and the text information cannot be fully represented. This paper used multi-scale feature extraction and fusion methods in the image feature characterization and text information representation sections of the VQA system, respectively to improve its accuracy. Firstly, aiming at the image feature representation problem, multi-scale feature extraction and fusion method were adopted, and the image features output of different network layers were extracted by a pre-trained deep neural network, and the optimal scheme of feature fusion method was found through experiments. Secondly, for the representation of sentences, a multi-scale feature method was introduced to characterize and fuse the word-level, phrase-level, and sentence-level features of sentences. Finally, the VQA model was improved using the multi-scale feature extraction and fusion method. The results show that the addition of multi-scale feature extraction and fusion improves the accuracy of the VQA model.

**Keywords** Multi-scale · Image features · Text information · Feature extraction and fusion · VQA

## 1 Introduction

VQA is a research task that links computer vision (CV) to natural language processing (NLP). As a high-level complex task, VQA not only includes the task of identifying and detecting the content of the picture, but also includes the task of understanding the problem and the research of combining the question with the picture, and the task of using the question and the picture content to reason about the answer. Therefore, a complete VQA model mainly includes three

steps: single-mode feature extraction, image feature and text information fusion, and answer reasoning [1–5]. Most of the early work applied the model of image caption generation [6–9] to the Visual Question Answering (VQA) task [10–12] and realized the ability of VQA by combining the modules of convolutional neural network and cyclic neural network. At present, there are several mainstream methods of VQA, such as the end-to-end neural network, the multi-module neural network, and the neural network with external database. The combination model is to decompose VQA tasks into independent models to solve different types of tasks, among which neural module networks (NMN) [13] and dynamic memory networks (DMN) [14] are typical representatives. Furthermore, scholars put forward the introduction of an external knowledge base [8, 15], among which the common external knowledge base includes the DBpedia [16], the Freebase [17], the YAGO [18], and the OpenIE [19].

The core task of the VQA model is the extraction and fusion of features [20, 21], which is also related to the accuracy of the model [22]. Because the features extracted by the deep neural network and then extracted by the shallow neural network have different meanings, the multiscale feature

✉ Mingzhe Liu  
liumz@cdut.edu.cn

✉ Wenfeng Zheng  
winfrms@uestc.edu.cn

<sup>1</sup> School of Automation, University of Electronic Science and Technology of China, Chengdu 610054, China

<sup>2</sup> School of Data Science and Artificial Intelligence, Wenzhou University of Technology, Wenzhou 325000, China

<sup>3</sup> College of Resource and Environment Engineering, Guizhou University, Guiyang 550025, China

<sup>4</sup> Department of Geography and Anthropology, Louisiana State University, Baton Rouge, LA 70803, USA

fusion method was applied to the extracted feature of the image and the text in the VQA system, and the joint complementation of the two can make the feature expression more perfect and more conducive to recognizing the object in the image.

### 1.1 Image Feature Extraction

There are two typical methods used in the task of image feature extraction. One of the typical deep learning methods for image feature extraction is region selection, which was proposed by Girshick [23] in the target detection task, and then widely used in image target detection and segmentation task. This method can get the object boundary and its key characteristics well, but the problem is that it will ignore some information, such as walls in the background, grass, etc. The second method uses the classical neural network, which has been pre-trained to extract features, such as the VGG model and the ResNet model trained in big data such as ImageNet. The advantage of this method is that it can use the existing classical model, which has been pre-trained by many data. On the other hand, the disadvantage of this method is that it requires that the new task data needs to be not much different from the data used in the pre-training. Otherwise, it will not achieve the desired effect.

The Convolutional neural network has made outstanding achievements in image classification [24, 25], but this method still has significant limitations. The most typical is that although the convolutional neural network is adaptive to the object's position change in the image, it is sensitive to the size change of the object. Therefore, if the object's size exceeds the convolutional kernel receptive field or the object's size is too small, the network cannot correctly identify the object from the image. Given this situation, scholars have proposed many solutions, such as image pyramid [26] or feature pyramid [6].

#### (1) Image Pyramid

The most intuitive method is to change the resolution of the image. In the earliest method [26], some scholars proposed to zoom the original image to different sizes to get an image pyramid structure, as shown in Fig. 1. After that, the

neural network used for classification is applied to images of various sizes. To some extent, this method solves the problem that the neural network is sensitive to the size of the target, but the cost is vast. Moreover, it needs to consume several times more storage space and computing resources than the general network [27], so this method is less used.

#### (2) Feature Pyramid

In the later improved methods, some scholars focus on the information representation of image features. The research shows that the shallow image feature has a small receptive field, high resolution, and rich location information, which is beneficial for detecting small objects. While the high-level feature has a large receptive field and a low resolution, strong semantic information representation complements shallow network information. So scholars put forward the SSD (single shot multi-box detector) [6] method (Fig. 2). In this way, the shallow network can detect some objects that the original deep network features cannot detect. However, the effect is still limited because of the lack of semantic information of the shallow network features.

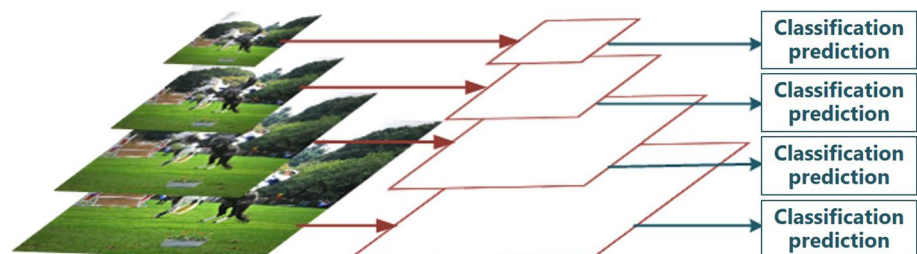
#### (3) Feature Pyramid Network

In order to combine the high-resolution features of shallow networks with the high-level semantic information of high-level network features, the feature pyramid network (FPN) method is proposed (Fig. 3). It extracts the features from the deep-layer network and gets the same features as the shallow layer features through the up-sampling. Then, the fusion of the two features is used as the input of the classification network and achieves good results in the classification task.

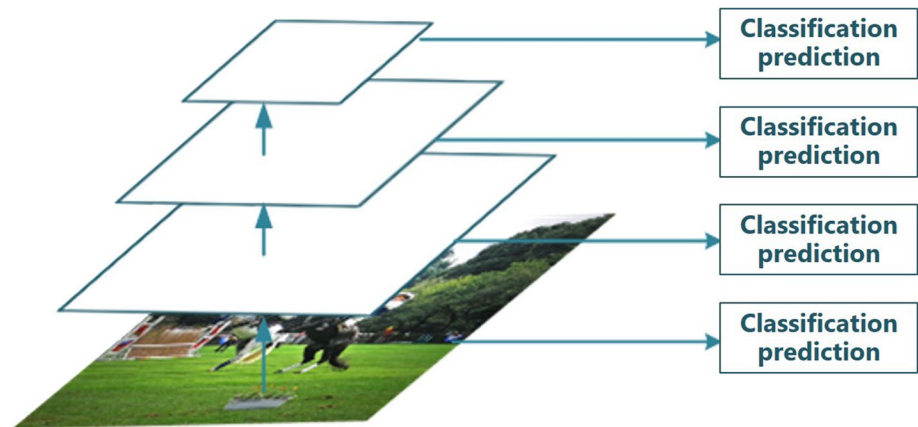
### 1.2 Text Information Extraction

The text information in the VQA database is unstructured information that is difficult for computers. To make the computer be able to deal with the text information, it needs to be transformed. Word embedding was used to map text to feature space. There are two models of word embedding: skip-gram model and continuous word bag mode (CBOW)

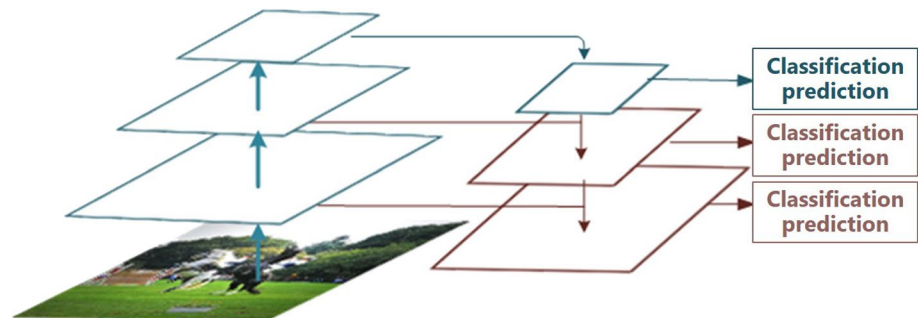
**Fig. 1** Pattern diagram of image pyramid



**Fig. 2** Pattern diagram of feature pyramid



**Fig. 3** Pattern diagram of feature pyramid network



[28]. In the skip-gram model, the model trains the neural network to derive the probability distribution of the context vector of the input word based on the given input word and the pre-sized window. CBOW, on the other hand, is given context and trained on input words, and the context under the window size given by the input predicts the probability distribution of words.

For the feature extraction of the question sentence, because the convolution neural network cannot process the sequence information well, the cyclic convolution neural network is used to get the sentence feature after iterating the word feature in the sentence. However, gradients in cyclic convolutional networks disappear as iterations increase. The long short-term memory (LSTM) network and the Gate Recurrent Unit (GRU) network [29] are proposed for this issue. LSTM uses gate control to control the time length of the self-accumulating information cycle. The sigmoid activation function limits the value to a limited range, which keeps the memory of the past gradient within a particular time and guarantees that the gradient will not explode due to accumulation. On the other hand, GRU simplifies the threshold and only uses the update gate and reset gate to achieve a similar effect with LSTM.

The realization of multiscale text feature extraction is based on the processing of language feature sequences. After embedding words into natural language, the obtained word

vectors are the features at the word scale. To get the feature in phrase scale, a one-dimensional convolution neural network proposed by Lu [30] is used to extract the feature of word vector space. Some scholars [31] propose using convolution kernels of three different scale windows to convolute with word vectors, respectively to get text features of different scales, cascade the output under all windows, and use the maximum pooling processing to get the “phrase” feature.

### 1.3 Multiscale Feature Fusion

In the VQA task, the typical multimodal features are image features and text features [32–34]. The fusion of image features and text features in the VQA system plays a vital role in model performance [35]. The selection of methods directly affects the results. In the early stage of the VQA model, the joint representation method is used to process multimodal features. Researchers map image features into text feature space, add features, measure similarity or join features in-depth at the element level [36, 37]. The fusion of semantic and image features adopts the direct connection method, which has an unclear termination position. Malinowski [10] proposed a recursive neural network implementation using long-term and short-term memory cells. The research of fine-grained classification tasks inspires the application of collaborative representation in multimodal

feature fusion. Lin first proposed the bilinear network [38], mainly used for fine-grained image classification tasks, such as distinguishing bird species and aircraft species and other extremely subtle images.

Because the feature dimension of the bilinear model is too high and there are many parameters, many scholars have proposed to compress and reduce the dimension of the fused feature. Gao [11] proposed compressing the bilinear pooling network, which reduced the feature dimension to a certain extent. Fukui [39] introduced a bilinear pooling network into a VQA task (MCB). In the VQA task, two independent convolutional neural networks of the original bilinear network are replaced by a convolutional neural network to extract image features and a cyclic neural network to extract text features. Both image and text features are mapped to the same feature space, and the subsequent feature fusion is still the same as Gao's processing. Although the method of MCB can compress the vector to a lower dimension, in order to ensure the classification result, the output feature is still a feature in a higher dimension space. Next, Yu [40] put forward the multi-modular factorized bilinear pooling (MFB) structure based on the MLB pooling method to reduce the dimension. The author mainly decomposes the feature mapping matrix into two low-rank matrices, making the feature mapping complete under relatively low features. Kim [41] introduced Hadamard product and matrix decomposition to compress the features. Ben [42] introduced Tucker decomposition of a tensor into a bilinear pooling network, further compressed data. Bilinear pooling text keyword features extracted by cyclic convolution neural network and feature output of convolution neural network and used text features and visual features to predict attention weighting in feature space.

This paper introduces the multiscale feature technology into the VQA system, and the improved image multiscale feature extraction method is introduced. Firstly, the development and research significance of multiscale feature fusion technology were introduced. After that, this paper introduces the multiscale image feature extraction method and the text multiscale feature extraction method. Finally, the experiments on the improved image multiscale feature method and the introduced text multiscale feature method are carried out and analyze its influence and role in the VQA task.

The main contributions and innovations are: apply multiscale feature fusion method to feature extraction of image and text in visual question answering system, and improve the multi-scale image feature representation method of image. Because the features extracted by the deep network and the shallow network in the neural network have different meanings, the combination of the two can make the feature expression more perfect and more conducive to the recognition of objects in the image. This paper introduces the

multi-scale feature extraction based on the depth residual network. Due to the existence of multimodal feature fusion based on bilinear network in the VQA model, the parameters of neural network are complex and huge. In this paper, the image multi-scale feature technology is improved, and the feature expression is simplified on the basis of ensuring performance improvement.

## 2 Materials and Methods

### 2.1 Materials

#### 2.1.1 Dataset

VQA dataset is one of the earliest and most widely used data sets. It contains VQA data set composed of real natural scene photos and VQA abstract data set composed of cartoon pictures. VQA data contains 123287 photos as the training set and 81,434 photos as the test set. The source of the photos is the coco data set provided by Microsoft for the image classification competition. The data of questions and answers are put forward and answered manually. The forms of questions include yes or no questions, multiple choices questions, and open answer questions. VQA consists of 614163 Q & A pairs. The provider gives a detailed analysis of the characteristics of the dataset. This paper mainly analyzes the open problems.

#### 2.1.2 Experimental Environment

The hardware environment is as follows: processor, CPU Intel i7-4790; graphics card, NVIDIA GTX 1070; memory, 16G. The Software environment is as follows: general computing architecture, CUDA 9.0; GPU acceleration library, CUDNN 7.0; deep learning framework, python.

### 2.2 Method

#### 2.2.1 Multiscale Image Feature Extraction and Fusion

In our experiment, the pre-trained residual network is used for image feature extraction. Since the visual question and answer database adopted in this paper is based on the COCO image data set, the pre-trained Resnet-152 model in the standard model library can be used as the image feature extraction network. Thus, this paper extracted the final output of conv3, conv4, conv5 residual modules. Regarding the fusion of multiscale features, we used the method of the FPN model to fuse the combination of features of different layers in a top-down way.



## 1. Image Feature Extraction Network

Image feature extraction adopts the feature extraction method based on deep learning. First, this paper removes the fully connected layer, which is highly related to the classification task. Then, the output of the final convolution layer is adopted as the image feature. Finally, the 152-layer residual network, pre-trained in ImageNet, is adopted for the image feature extraction model.

## 2. Feature Extraction Network Fine-Tuning

After considering the difference in training data distribution from the original model, the network is fine-tuned. He Keming [43] studied the parameter initialization method of ImageNet pre-training. The research shows that the advantage of this method is that it can improve the convergence speed but is not sensitive enough to local targets. Unlike ImageNet, which was originally only used for classification tasks, the COCO data set contains more multi-objects

data and more abundant local targets. Therefore, this paper fine-tuned the network on the COCO data set.

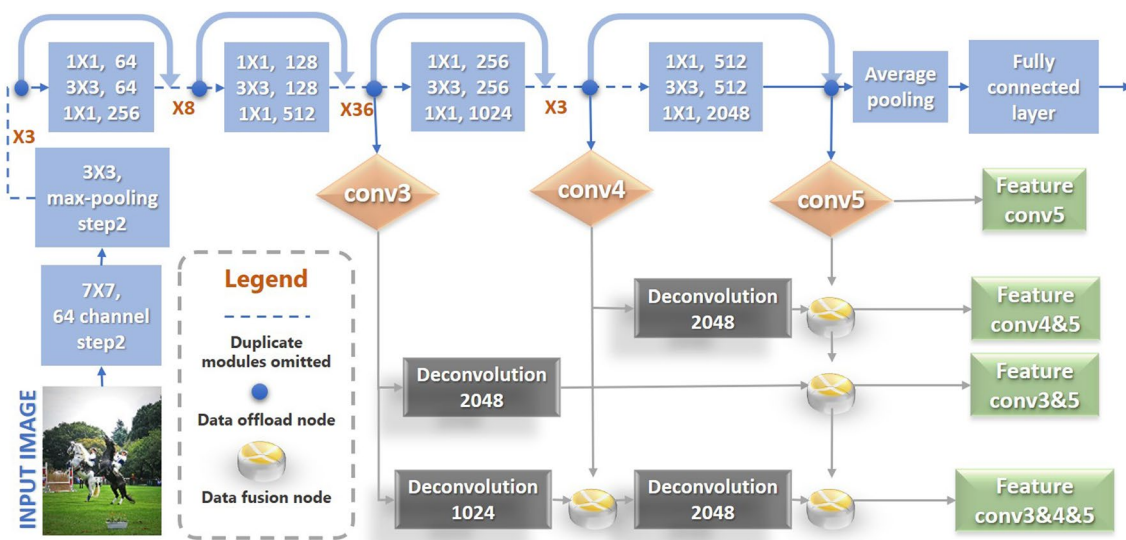
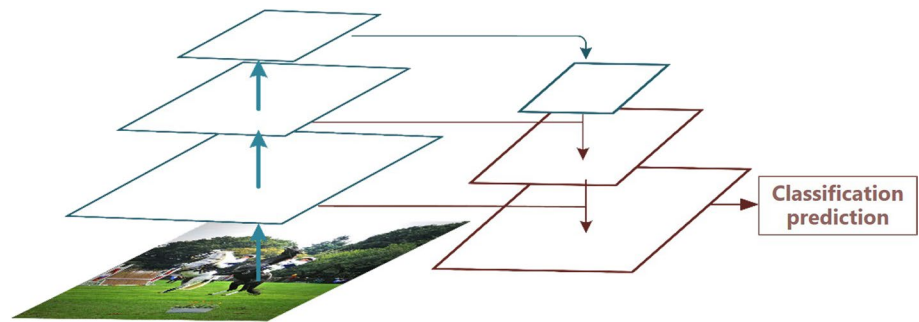
## 3. Multiscale Feature Extraction and Fusion

The bilinear feature fusion networks generate a large number of parameters with high dimensions. Therefore, considering the computational cost, instead of generating answers to all scale features, the experiment explores the best fusion mode among the features of different scales and finally obtained a single fusion feature, as shown in Fig. 4.

First, we unify the size of the images in the COCO data set to  $448 \times 448$  as the model input. The output feature size of conv3 is  $28 \times 28$ , and the channel dimension is 512. The output feature size of conv4 is  $14 \times 14$ , and the channel dimension is 1024. The output feature size of conv5 is  $7 \times 7$ , and the channel dimension is 2048, shown in Fig. 5.

We refer to the FPN fusion method and fuse different scale features in top-down order for feature fusion. For example, for the merge of conv5 with conv4, we first deal with features extracted by conv5. Then, this paper adopts

**Fig. 4** Multi-scale feature fusion in visual question answering system



**Fig. 5** Multi-scale feature extraction model for images in VQA system

the 1024 times deconvolution, in which step length is 2 and dimension is 2048, expand the feature to the same feature size and channel number as conv4. Then we add the features by elements, and the new features are stored as conv4 & 5. The conv5 and conv3 fusion are similar approaches.

Then, for the integration of three layers of features, we further deconvolution conv4&5. With 512 deconvolution and step of 2, the size of features is expanded to  $28 \times 28$ , and the number of channels is reduced to 512, which is the same as the feature size and channel number of conv3. After the expansion, conv4&5 and conv3 are also processed by adding elements to get the high-resolution features of conv3&4&5. As the experimental control, we retain the direct output of conv5 as the final feature directly outputted by the model.

#### 4. Feature Fusion Networks Applied to Different Multimodal

The proposed multiscale feature extraction and fusion method are applied to different multi-mode feature fusion networks. The experiment adopts the MLB [2] network and the MUTAN [44] network and compares the results after experiments.

#### 5. Generate the Answer

A multilayer perceptron is employed to classify the fusion features, and the probability distribution of the final answer is obtained. Finally, the features are decoded into natural language by the circular convolutional neural network.

#### 2.2.2 Multiscale Text Feature Extraction

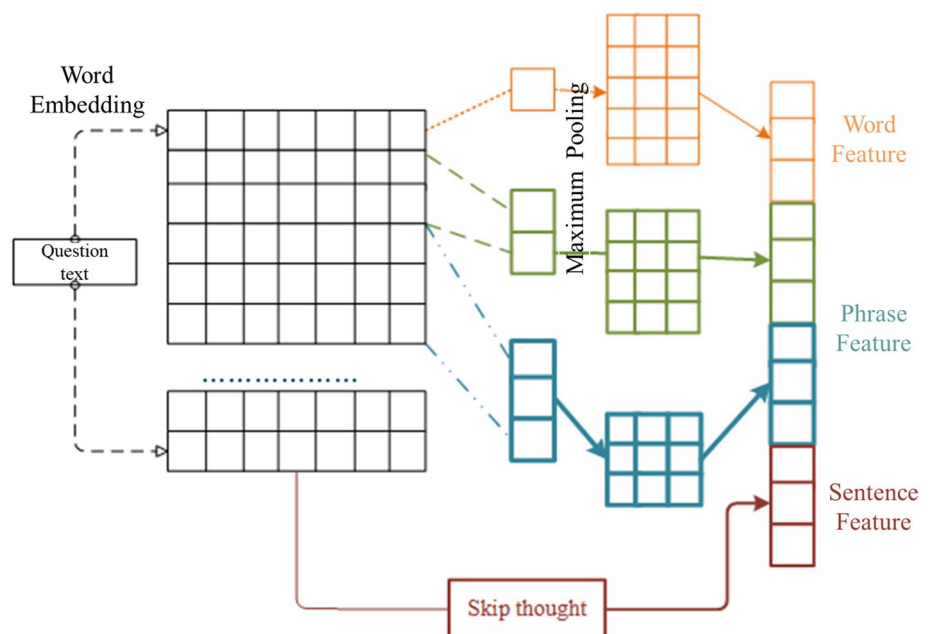
In the VQA model, we first embed words into the text and mapped it to the continuous numerical feature space to get the continuous word vector. Then the phrase features are obtained by one-dimensional convolution with three different scales. Moreover, the skip-thought model is adopted for feature extraction of sentences. Combined with the expression of GRU, the word embedding feature of a single step went through convolution in the unit of window scale and then output as word feature. Sentences with the length of  $N$  are filled to the set input length and output as sentence features by skip-thought. For phrase features, maximum pooling output after features is convoluted with a different scale one-dimensional convolution network. The output is phrase features. The overall text feature extraction is shown in Fig. 6.

As the skip-thought loop network is an encoder, we need to set the length of the hidden layer first and set the target cycle times as length  $N$ . Sentences with insufficient length are filled with 0. The word features and phrase features are obtained after embedding with the one-dimensional convolution of different scales. For phrase feature extraction, the experiment adopts a convolution kernel of various sizes and a maximum pooling layer to extract the feature of word vector as phrase feature.

#### 2.2.3 Overall Model

The model is divided into two parts, as shown in Fig. 7. Firstly, the image multiscale feature extraction and text multiscale feature extraction modules described extract the

**Fig. 6** Multiscale text feature extraction



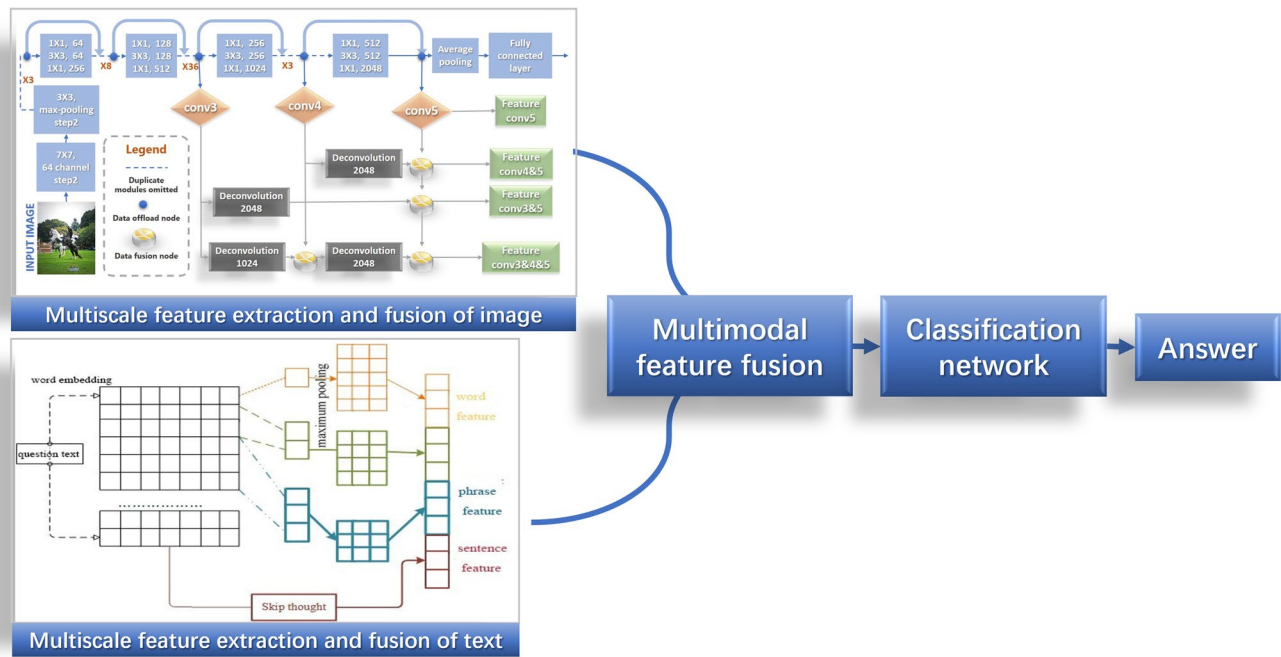


Fig. 7 Overall model

features of VQA data. Then, the extracted features are converted into binary data and stored on the hard disk. This section sets up the VQA system model, the ResNet152 network, which has been pre-trained in the ImageNet data set, is employed to feature image extraction. The feature extraction model of the problem employs skip—thought cyclic convolution neural network based on the structure of the GRU helped threshold. The later part is to complete feature fusion, interaction, and the formation of the answer.

The fusion method needs to be combined with the tensor compression method mentioned above. This paper uses the bilinear network to realize the fusion of the final output of multiscale text and image features. After feature fusion, we use MLB and tucker decomposition to compress the feature. The compressed features are input into the answer to generate multiple classifiers, and the probability distribution of each answer is obtained. Finally, the final answer is obtained by fusing and normalizing it with the softmax layer.

After the model features are extracted, we store the features as binaries for subsequent question-and-answer system training. In the question-answering system, the fused multiscale image features and text features are used as the input of the question-answering system, and the value of state distribution is firstly used for parameter initialization. After a few times of training, the parameters and results of each training are stored. After each training, the parameters with the best results were selected for initialization. In this

paper, the Adam optimization algorithm was used for the training optimizer. The Alpha value was set to 0.01, the first-order moment was estimated to be 0.9, and the second-order moment was estimated to be 0.99.

### 3 Results

Firstly, the comparison of several different fusion modes is given. Then, the fusion mode with better performance is selected as the basis of the subsequent work to be applied in the model. Next, multiscale feature fusion is applied to the model, and the method is analyzed after the experiment.

#### 3.1 Comparison of Fusion Methods

The experiment tests several visual question-and-answer models at the current forefront, among which Baseline is the original image feature and text feature cascade model, as shown in Table 1.

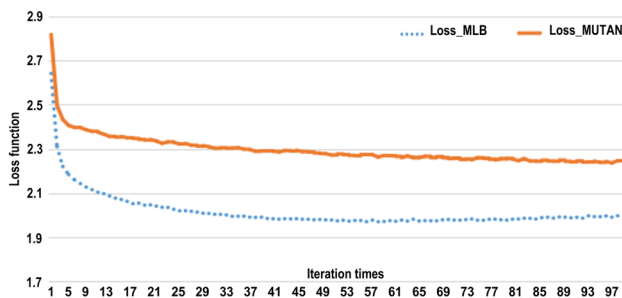
In this experiment, MLB fusion and the MUTAN fusion are used as the control group to compare the prediction results of MLB fusion and the MUTAN fusion on the open problem data set in the VQA data set. The loss function and the accuracy under top1 and top5 indexes are analyzed and compared, as shown in Table 2. Top1 refers to the model as successful learning only when the answer with the highest score is correct in the probability distribution obtained

**Table 1** Experimental results of various VQA models

| Model    | Accuracy (%)   |          |
|----------|----------------|----------|
|          | Validation set | Test set |
| Baseline | 67.92          | 70.73    |
| MCB      | 70.74          | 73.18    |
| MLB      | 71.53          | 74.29    |
| MUTAN    | 71.72          | 73.89    |

**Table 2** Comparison of MLB model and MUTAN model

|      | MLB_train (%) | MLB_test (%) | MUTAN_train (%) | MUTAN_test (%) |
|------|---------------|--------------|-----------------|----------------|
| Top1 | 59.28         | 47.28        | 54.47           | 47.67          |
| Top5 | 82.97         | 74.29        | 79.20           | 73.89          |

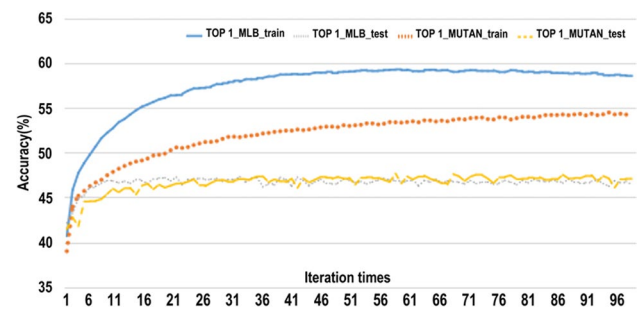
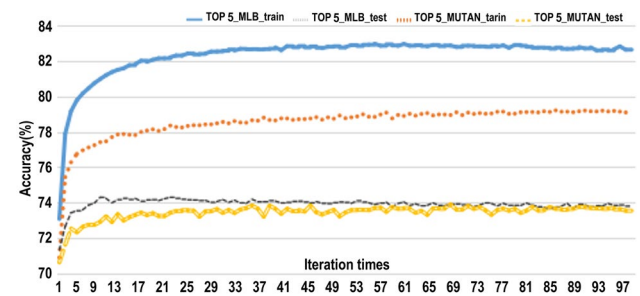
**Fig. 8** Loss function

through training. Top5 refers to the obtained probability distribution. Therefore, the model learning is considered successful when the highest five results contain correct answers.

According to the convergence rate of the model's loss function in Fig. 8, the model's loss function based on MLB fusion converges significantly faster in the training process. The model's loss function based on the MUTANMLB tensor dimension-reducing model converges to 1.98, and the final loss function of the MUTAN method converges to 2.27.

As shown in Fig. 9, under the evaluation of top1 accuracy, the highest accuracy of the MLB model on the training set was nearly 60%, which was higher than the 54% accuracy of the MUTAN method. The test results of the two were very similar under the analysis of the generalization effect of the test set.

As shown in Fig. 10, under the top5 evaluation method, the training set results still showed that the accuracy of the MLB model is higher than that of the MUTAN model, with

**Fig. 9** Accuracy of Top1 in test set and training set**Fig. 10** Top 5 accuracy of MLB and MUTAN in test set and training set**Table 3** Multi-scale text feature experiment

| Method                   | MLB (%) | MUTAN (%) |
|--------------------------|---------|-----------|
| Multi-scale text feature | 74.82   | 74.05     |
| Original model           | 74.39   | 73.89     |

a difference of nearly 4%. On the test set, the result of the MLB model is 0.4% higher than that of the MUTAN model, but the MLB model converges in the 10th epoch, while the MUTAN model converges in the 60th epoch.

### 3.2 Comparison of Multiscale Feature Fusion Methods

We selected MLB and MUTAN models as the multimodal feature fusion method of the VQA model by comparing and testing various models before, and firstly added text multi-scale features to conduct the experiment.

As shown in Table 3, the accuracy of the MLB model and MUTAN model was improved to some extent after adding multiscale text features, among which MLB improved by 0.43 percentage points.

Based on multiscale text feature extraction, we experiment with the multiscale image feature method. Multiscale feature extraction is based on the image FPN network



**Table 4** Image feature combination experiment

| Combination           | MLB (%) | MUTAN (%) |
|-----------------------|---------|-----------|
| Conv5                 | 74.82   | 74.05     |
| Conv5 + conv4 + conv3 | 73.34   | 72.03     |
| Conv5 + conv4         | 74.50   | 74.93     |
| Conv5 + conv3         | 75.97   | 75.88     |

method. In the calculation, the FPN network synthesis consumes more computation resources and time. In our experiment, each characteristics layer is in different combinations, and we analyzed the result of the combination to get a better solution.

Firstly, based on the above experiment, the directly extracted feature layer includes conv3, conv4, and conv5. The obtained accuracy when the input is the feature of conv5 is taken as the benchmark. Combined conv3 and conv5, conv4 and conv5 are taken as the input, respectively to calculate the accuracy, and then the combined three features are taken for the experiment. It can be seen from Table 4 that when using three layers of features with different scales at the same time, the results decreased. After combining conv4 and conv5 features, the experimental results did not significantly improve. After combining conv3 and conv5 features, the experimental results improved significantly, and MLB and MUTAN both increased by about two percent under the condition of top5 evaluation on the test set.

## 4 Discussion

Table 1 show that using dual linear network test results of multiple modal characteristics of fusion is superior to the traditional method (Baseline) test results. Furthermore, the MLB model and MUTAN model are better than the original bilinear model of MCB. Although these two models are processed reduction of dimensionality, the parameter number is less than MCB mode.

Table 2 (Comparison of MLB model and MUTAN model) shows the maximum accuracy of training 100 epochs. The average results under the training set show that the training accuracy of the MLB fusion in the training data set is significantly higher than that of the MUTAN model Tucker decomposition method. However, there is little difference between the two in the test set. There is only a 0.39% difference in the top1 accuracy rate in the test set and only a 0.4% difference in the top5 accuracy rate. The difference under top1 evaluation in training test is nearly 5%, and the difference under top5 evaluation is 3.77%. Thus, the MUTAN tensor compression model has a relatively similar effect compared with the MLB model. However, the MUTAN model reduces

more parameters and feature dimensions and is better than the MLB model in memory consumption.

From Fig. 10 (Top5 accuracy), it can be seen from the analysis that the MLB model converged faster, and the effect was slightly better than the MUTAN model. However, the overfitting of the MLB model is more severe than that of the MUTAN model. The reason may be that the MLB model is more complex and has more parameters. Therefore, MUTAN is better at solving the overfitting problem than the MLB model, but the training effect of the model is not improved. Therefore, considering that there is little difference between the two methods in the performance of final results, we can adopt the two methods in the follow-up research and further compare them in the follow-up work.

From experimental results analysis, we find that when the feature fusion of the three layers of different scales is used simultaneously, the results did not improve but declined. After combining the feature of conv4 and conv5, the improvement is not obvious. However, after combining the feature of conv3 and conv5, experimental results show evident improvement. On the test set based on the top5 evaluation, the improvement under the MLB and MUTAN model are both about two percent.

From the extracted features, we can see that the image features under different scales have obvious differences, representing different aspects of the image information. Therefore, this information can be combined with the problem features under different scales in future work.

## 5 Conclusions

This paper improves the image feature extraction method in the VQA model. The concept and method of multiscale features are proposed in the research of feature engineering. The application of multiscale features in image classification tasks has achieved good results. Refer to the semantic research of different scale features in feature engineering and the semantic information of different languages represented by different scale features of image features. This paper applies multiscale feature extraction and fusion in the VQA system. It also adopts a simplified multiscale feature method to integrate information of different scales and reduce the parameter number. The 152-layer ResNet network pre-trained on the ImageNet dataset is used for image feature extraction. The final output of the different residual convolution modules of the network is used as image features of different scales. In this paper, the characteristics of different scales are combined through experiments, and the best combination method is applied to the VQA system.

Although the work of this paper has made improvements to the VQA model in terms of visual feature extraction and text information extraction, the results still have not achieved

the ideal effect, and there is room for improvement and optimization.

First of all, when the model recognizes objects in the problem, it cannot accurately recognize and give answers in the image. Many scholars have discussed the causes and solutions to this problem. The underlying cause may be that there are many types of objects in the image. For example, objects occlude each other or objects are too small. At present, the possible solutions can refer to the fine-grained classification problem to perform separate target detection and classification of image features. However, the existing problem is the same as the dilemma faced by the fine-grained classification problem. Moreover, it will take up a lot of computing resources and storage space. Therefore, the solution to this problem requires further research on optimizing feature compression and dimensionality reduction.

**Acknowledgements** A thanks to all the authors for their great contributions to this paper.

**Author Contributions** Conceptualization, WZ and ML; methodology, ZY; software, YD and SL; formal analysis, LY and ML; data curation, YD; writing—original draft preparation, SL and LY; writing review and editing, WZ, LY and SL; visualization, YD; supervision, ZY and ML; funding acquisition, WZ. All authors have read and agreed to the published version of the manuscript.

**Funding** This work was jointly supported by the Sichuan Science and Technology Program (2021YFQ0003).

**Data Availability** Not applicable.

## Declarations

**Conflict of Interest** The authors have no competing interests to declare that are relevant to the content of this article.

**Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

## References

- Zhang, D., Cao, R., Wu, S.: Information fusion in visual question answering: a survey. *Inform. Fusion* **52**, 268–280 (2019). <https://doi.org/10.1016/j.inffus.2019.03.005>
- Kim, J.H., et al., Hadamard Product for Low-rank Bilinear Pooling. (2016)
- Chen, C., Han, D., Chang, C.C.: CAAN: context-aware attention network for visual question answering. *Pattern Recogn.* (2022). <https://doi.org/10.1016/j.patcog.2022.108980>
- Chen, Y.-S., Jiang, I.H.-R.: Many-layer hotspot detection by layer-attended visual question answering. In 25th design, automation and test in europe conference and exhibition (DATE). *Electr. Network.* (2022)
- Guo, Z., Han, D.: Multi-modal co-attention relation networks for visual question answering. *Visual Computer* (2022). <https://doi.org/10.1007/s00371-022-02695-9>
- Wu, Q., et al. What value do explicit high level concepts have in vision to language problems? in *Proceedings of the IEEE conference on computer vision and pattern recognition.* (2016)
- Yao, T., et al. Boosting image captioning with attributes. In *Proceedings of the IEEE international conference on computer vision.* (2017)
- Donahue, J., et al., Long-term recurrent convolutional networks for visual recognition and description. in *Proceedings of the IEEE conference on computer vision and pattern recognition.* (2015)
- Vinyals, O., et al. Show and tell: A neural image caption generator. In *Proceedings of the IEEE conference on computer vision and pattern recognition.* (2015)
- Malinowski, M., M. Rohrbach, M. Fritz.: Ask your neurons: A neural-based approach to answering questions about images. In *Proceedings of the IEEE international conference on computer vision.* (2015)
- Gao, H., et al.: Are you talking to a machine? Dataset and methods for multilingual image question. *Adv. Neural. Inf. Process. Syst.* **28**, 2296–2304 (2015)
- Ren, M., Kiros, R., Zemel, R.: Exploring models and data for image question answering. *Adv. Neural. Inf. Process. Syst.* **28**, 2953–2961 (2015)
- Andreas, J., et al. Neural module networks. in *Proceedings of the IEEE conference on computer vision and pattern recognition.* (2016)
- Kumar, A., et al., Ask me anything: Dynamic memory networks for natural language processing. In *International conference on machine learning.* PMLR. (2016)
- Xiao, F.Y., Pedrycz, W.: Negation of the quantum mass function for multisource quantum information fusion with its application to pattern classification. *IEEE Trans. Pattern Anal. Mach. Intell.* **45**(2), 2054–2070 (2023). <https://doi.org/10.1109/tpami.2022.3167045>
- Auer, S., et al. DBpedia: A nucleus for a web of open data. in *6th International Semantic Web Conference/2nd Asian Semantic Web Conference (ISWC 2007/ASWC 2007).* Busan, South Korea: Springer-Verlag Berlin. (2007)
- Bollacker, K., et al. Freebase: a collaboratively created graph database for structuring human knowledge. In *Proceedings of the 2008 ACM SIGMOD international conference on Management of data.* (2008)
- Hoffart, J., et al.: YAGO2: a spatially and temporally enhanced knowledge base from Wikipedia. *Artif. Intell.* **194**, 28–61 (2013)
- Etzioni, O., et al. Open information extraction: The second generation. In *Twenty-Second International Joint Conference on Artificial Intelligence.* (2011)
- Zhang, M., et al.: Multimodal feature fusion and exploitation with dual learning and reinforcement learning for recipe generation. *Appl. Soft Comput.* (2022). <https://doi.org/10.1016/j.asoc.2022.109281>
- Huang, W.-C., et al.: Attentive gated graph sequence neural network-based time-series information fusion for financial trading. *Inform. Fusion* **91**, 261–276 (2023). <https://doi.org/10.1016/j.inffus.2022.10.006>

22. Xiao, F.: GEJS: a generalized evidential divergence measure for multisource information fusion. *IEEE Trans. Syst. Man Cybern: Syst.* (2022). <https://doi.org/10.1109/TSMC.2022.3211498>
23. Girshick, R., et al.: Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. (2014)
24. Xiao, F., Wen, J., Pedrycz, W.: Generalized divergence-based decision making method with an application to pattern classification. *IEEE Trans Knowl Data Eng* (2022). <https://doi.org/10.1109/TKDE.2022.3177896>
25. Xiao, F., Cao, Z., Lin, C.T.: A complex weighted discounting multisource information fusion with its application in pattern classification. *IEEE Trans Knowl Data Eng* (2022). <https://doi.org/10.1109/TKDE.2022.3206871>
26. Adelson, E.H., et al.: Pyramid methods in image processing. *RCA Eng* **29**(6), 33–41 (1984)
27. Ni, X., et al.: Semantic representation for visual reasoning. *MATEC Web Conf.* **277**, 02006 (2019)
28. Carrasco, R.S.M., Sicilia, M.-A.: Sicilia, and security unsupervised intrusion detection through skip-gram models of network behavior. *Comput. Secur.* **78**, 187–197 (2018)
29. Chung, J., et al.: Gated feedback recurrent neural networks. In *international conference on machine learning*. PMLR. (2015)
30. Lu, J., et al.: Hierarchical question-image co-attention for visual question answering. *Adv. Neural. Inf. Process. Syst.* **29**, 289–297 (2016)
31. Nie, X., et al.: Deep multiscale fusion hashing for cross-modal retrieval. **31**(1), 401–410 (2020)
32. Miao, Y., et al.: Research on visual question answering based on GAT relational reasoning. *Neural Process. Lett.* **54**(2), 1435–1448 (2022). <https://doi.org/10.1007/s11063-021-10689-2>
33. Peng, L., et al.: Answer again: improving VQA with cascaded-answering model. *IEEE Trans. Knowl. Data Eng.* **34**(4), 1644–1655 (2022). <https://doi.org/10.1109/tkde.2020.2998805>
34. Xia, Q., et al.: Multi-modal alignment of visual question answering based on multi-hop attention mechanism. *Electronics* (2022). <https://doi.org/10.3390/electronics11111778>
35. Wu, Q., et al.: Visual question answering: a survey of methods and datasets. *Comput. Vis. Image Underst.* **163**, 21–40 (2017)
36. Miao, Y., et al.: Research on visual question answering based on dynamic memory network model of multiple attention mechanisms. *Sci Rep* (2022). <https://doi.org/10.1038/s41598-022-21149-9>
37. Yan, F., et al.: SPCA-Net: a based on spatial position relationship co-attention network for visual question answering. *Vis Comput* **38**(9–10), 3097–3108 (2022). <https://doi.org/10.1007/s00371-022-02524-z>
38. Lin, T.-Y., RoyChowdhury A., Maji S.: Bilinear cnn models for fine-grained visual recognition. In *Proceedings of the IEEE international conference on computer vision*. (2015)
39. Fukui, A., et al.: Multimodal compact bilinear pooling for visual question answering and visual grounding. In *conference on empirical methods in natural language processing*. ACL. (2016)
40. Yu, Z., et al.: Multi-modal factorized bilinear pooling with co-attention learning for visual question answering. In *Proceedings of the IEEE international conference on computer vision*. (2017)
41. Kim, J.-H., et al.: Multimodal residual learning for visual qa. In *Advances in neural information processing systems*. (2016)
42. Ben-Younes, H., et al.: Mutan: Multimodal tucker fusion for visual question answering. In *Proceedings of the IEEE international conference on computer vision*. (2017)
43. He, K., R. Girshick, P. Dollár. Rethinking imagenet pre-training. in *Proceedings of the IEEE/CVF International Conference on Computer Vision*. (2019)
44. Ben-younes, H., et al. MUTAN: Multimodal Tucker Fusion for Visual Question Answering. In *16th IEEE International Conference on Computer Vision (ICCV)*. Venice, ITALY: Ieee. (2017)

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.